

KOODAL

கூடல் where citizens and government come together

**Every civic issue, reported,
verified and resolved in
the open.**

• Team TRIOZ • Built with Gemini + Firebase



THE PROBLEM • INDIA'S CITIES

Reported. Forwarded. Forgotten.

Civic complaints go into a black box.



No visibility

Citizens never know if anyone saw it.



Duplicate noise

The same pothole, filed again and again.



Language barrier

Forms in English. People speak Tamil.



No proof of fix

"Closed" doesn't mean fixed.

THE SOLUTION

KOODAL closes the loop.

Citizens **report & verify**.

Gemini **organises**.

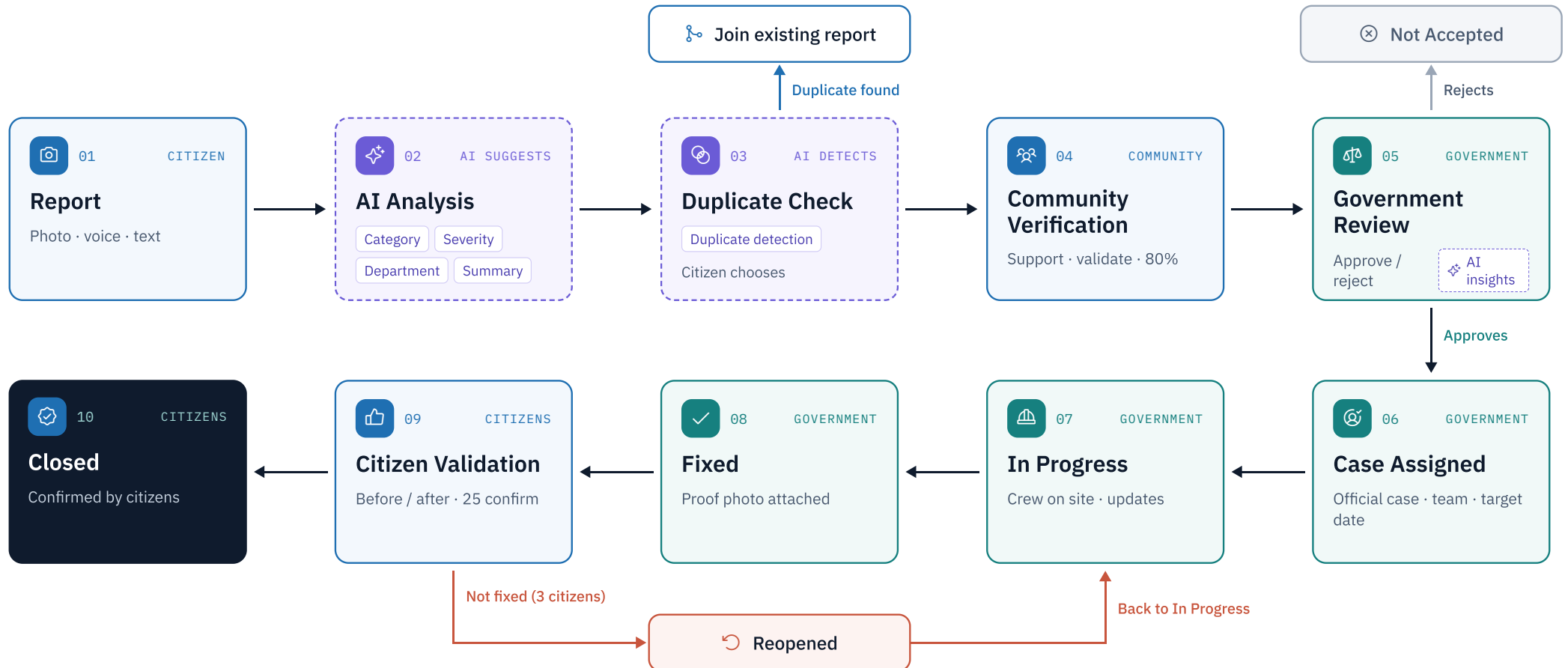
Government **resolves**.

Citizens **confirm**.



KOODAL — Complete Report to Resolution Flow

■ Citizens decide ■ Government decides ■ AI assists



✦ **AI suggests.** Citizens and government make every decision.

Gemini assists. People decide.

For citizens: speak, don't fill forms



வேளச்சேரி 100 அடி சாலையில் கழிவுநீர்
வழிகிறது

↓ Gemini structures the report

Category

Sewage & water

Severity

Critical

Department

Metrowater

Duplicate

92% match: join

For government: see what matters first



Priority score

Severity + confidence + support



Hotspots

Emerging issues on the map



Evidence summaries

Read a case in seconds

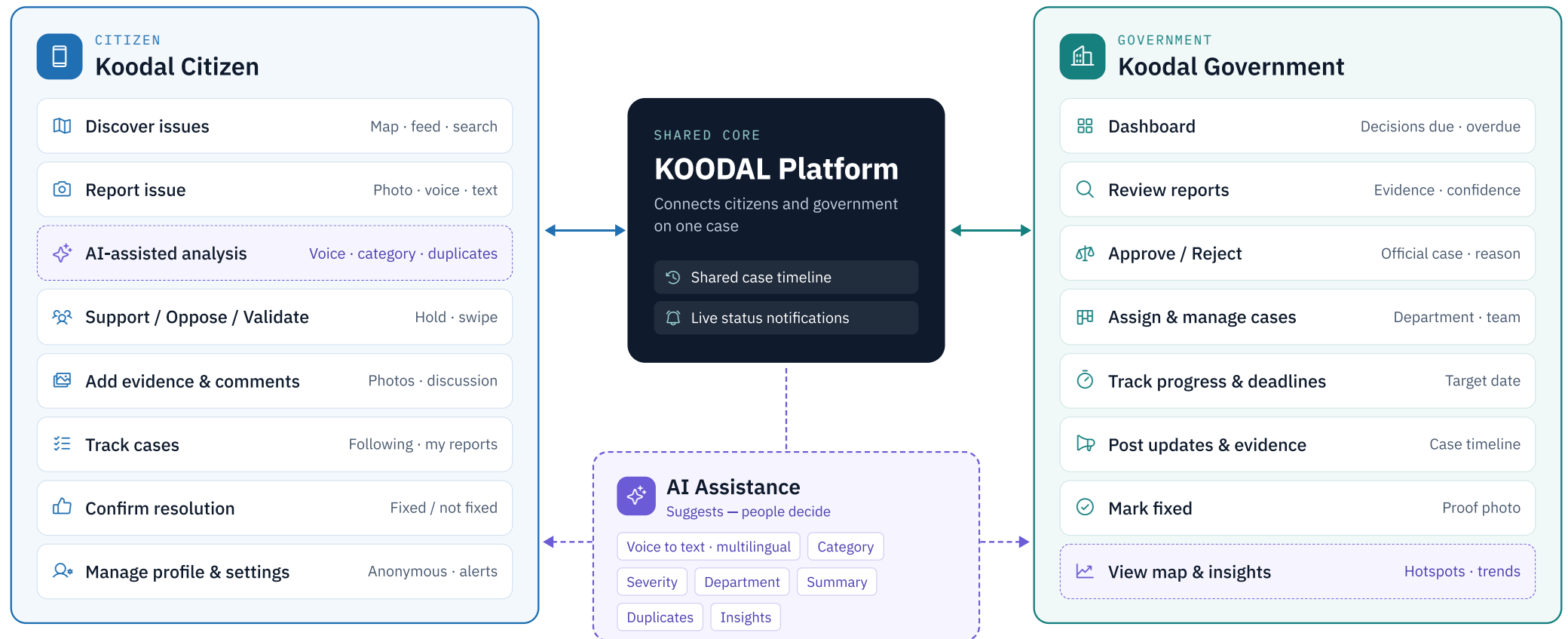


AI never approves, rejects or closes.

TWO ROLES · ONE PLATFORM

KOODAL — Platform Features by User Role

■ Citizen ■ Government ■ AI assists

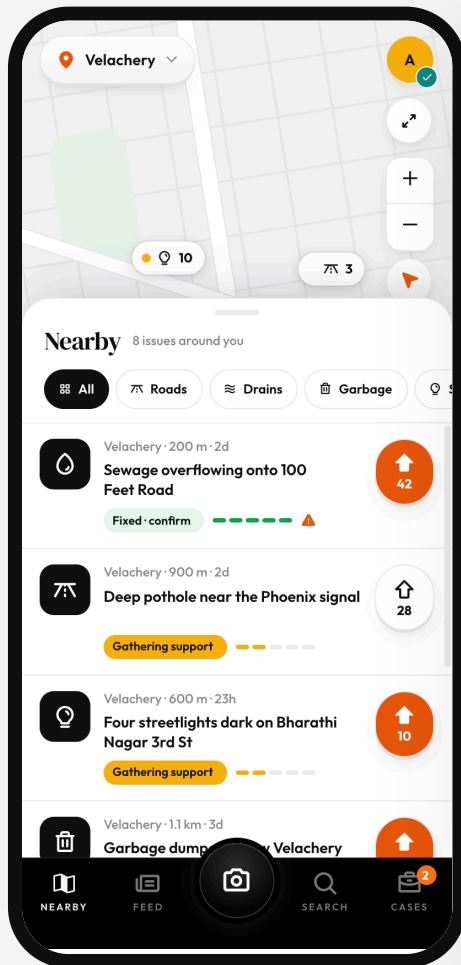


✦ **AI assists both roles.** Citizens verify and confirm, government approves and resolves.

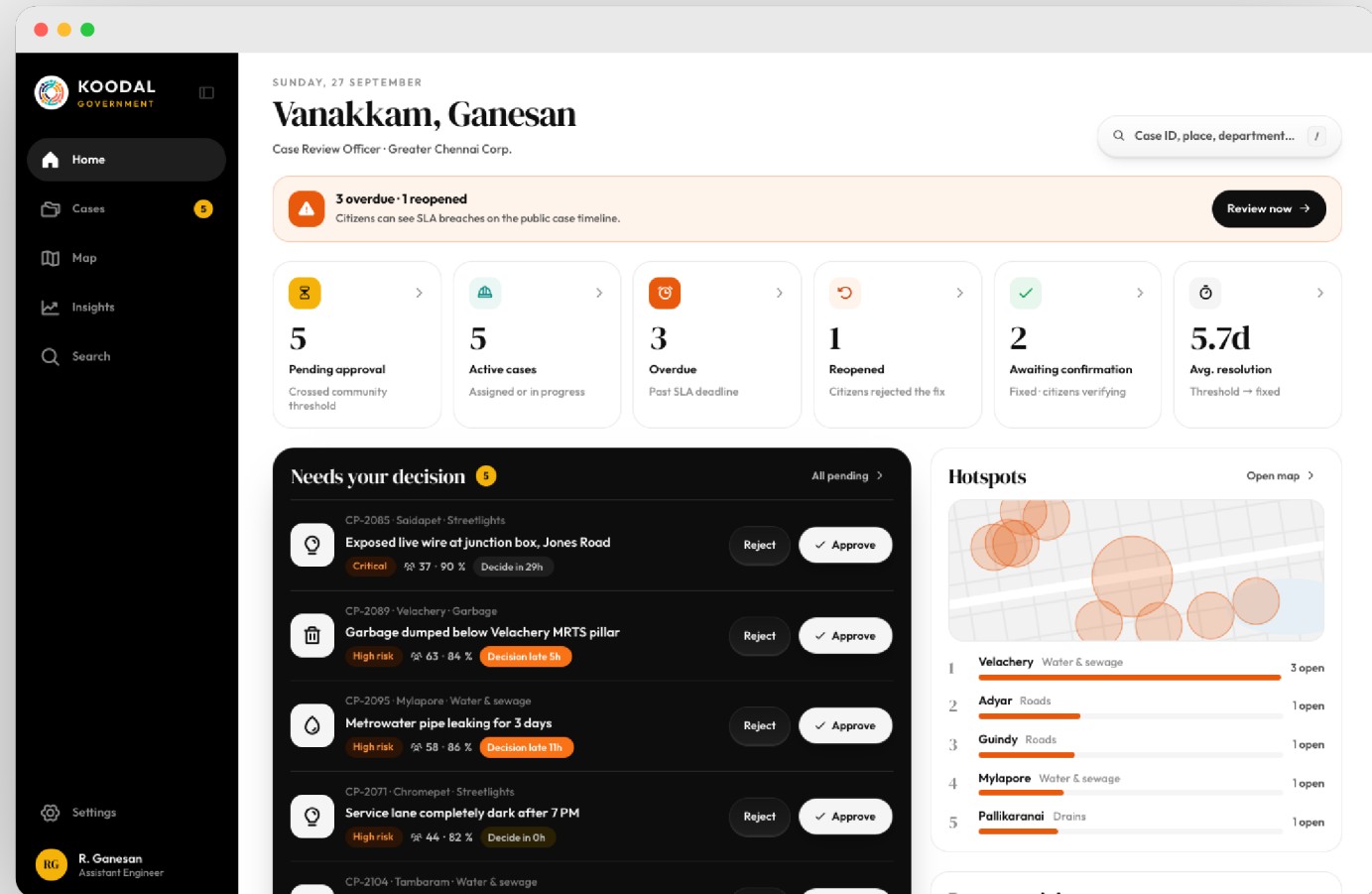
A working product.

● Same case data, both apps

Koodal Citizen • PWA

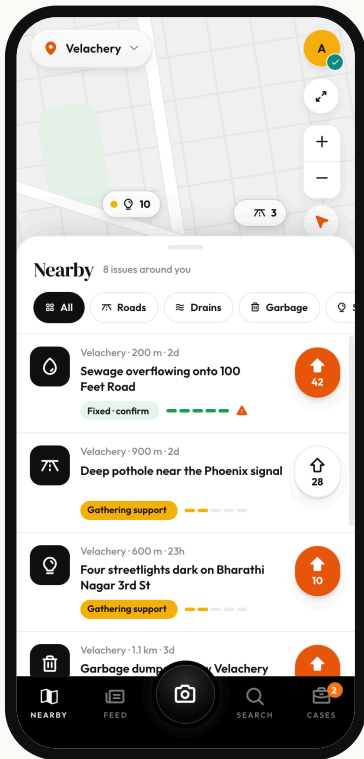


Koodal Government • Web console

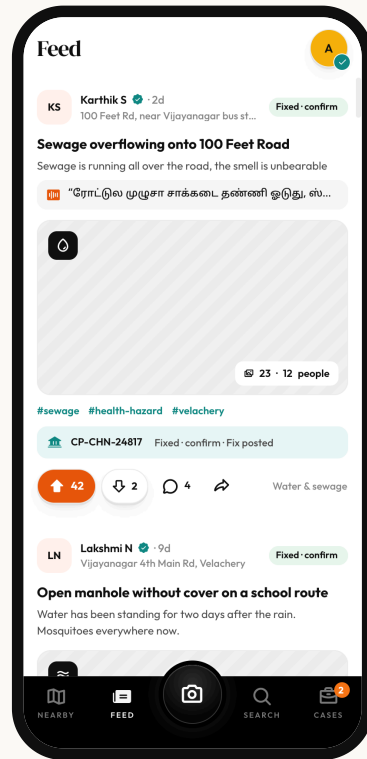


KOODAL CITIZEN · MOBILE PWA

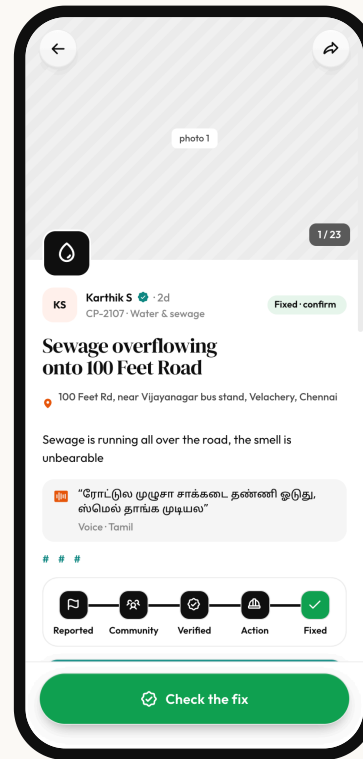
Report in seconds. Follow to the fix.



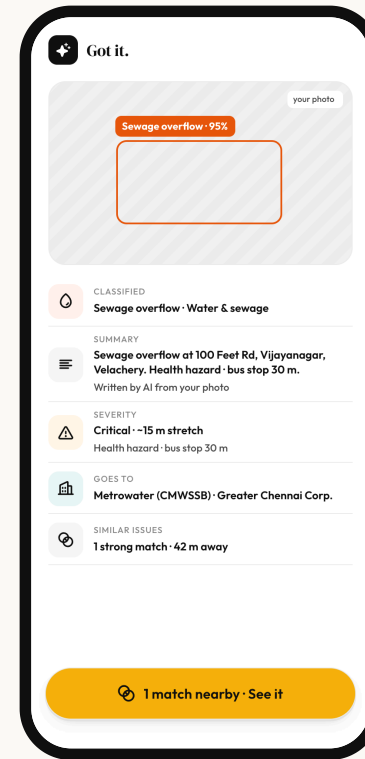
Nearby map
Issues around you



Community feed
Support · share



Issue detail
Hold to support



Gemini analysis
Category · duplicates

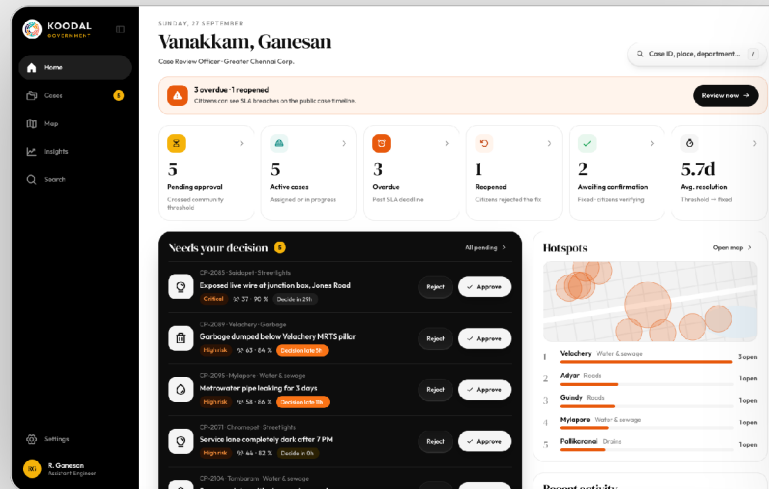


Confirm the fix
Before / after

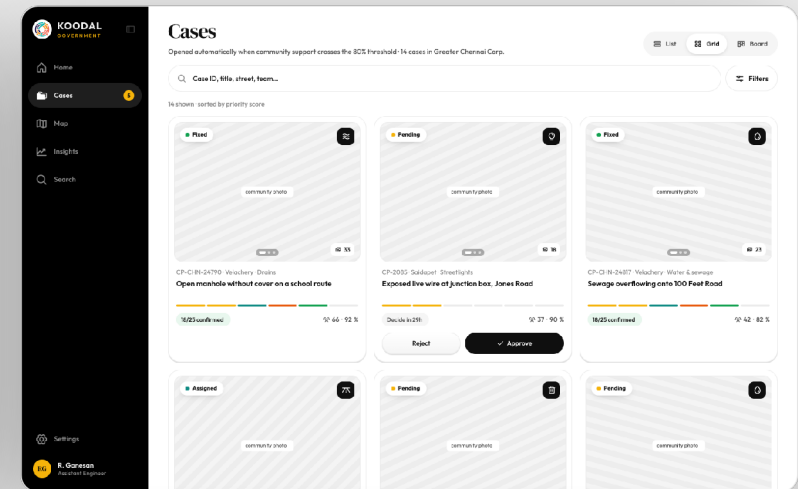
KOODAL
GOVERNMENT •
DESKTOP

Verified cases, clear deadlines.

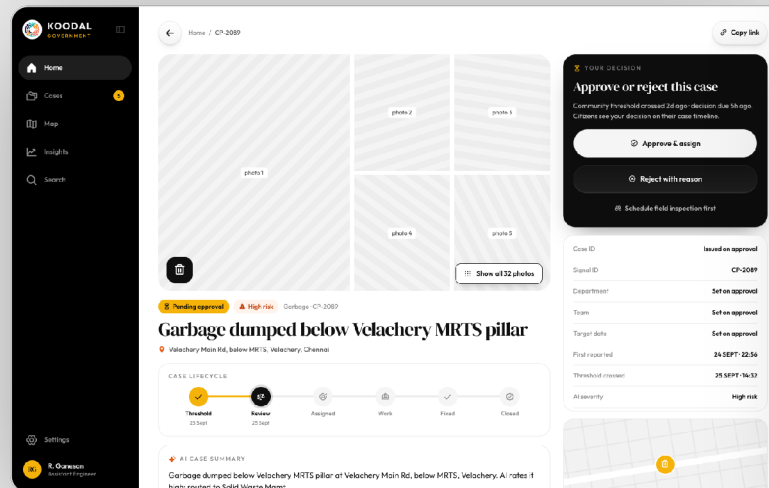
Officers only see reports
the community has already
verified.



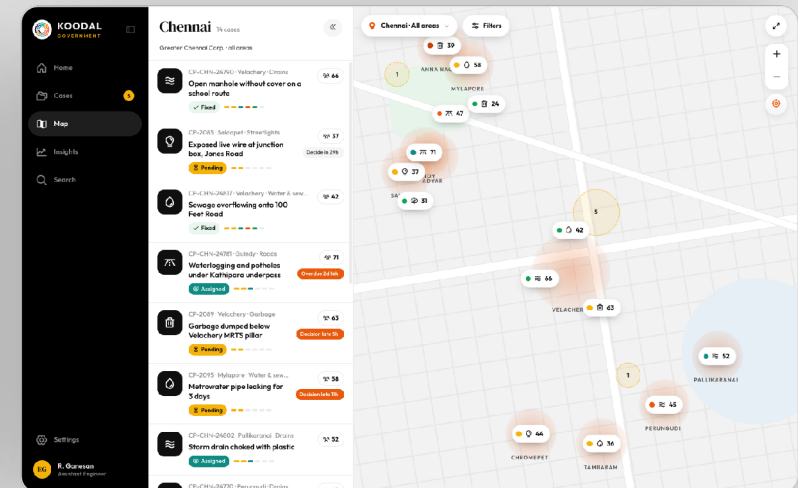
Dashboard Due · overdue · reopened



Case board Approve → fixed



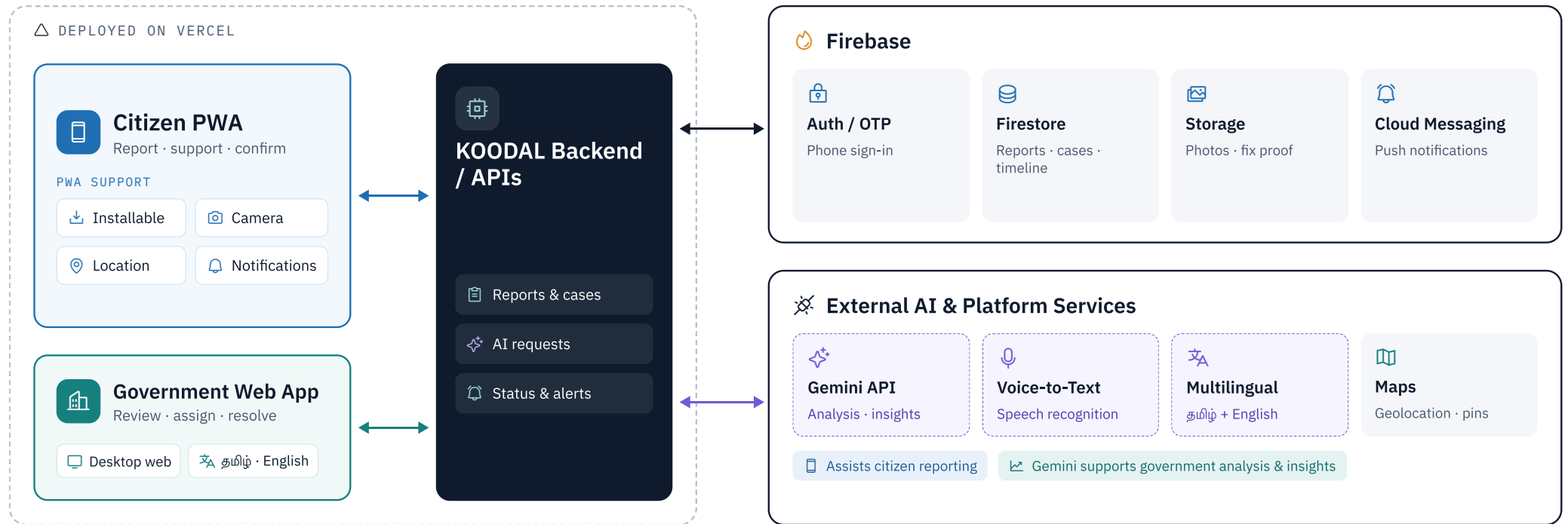
Case review Evidence + AI summary



City map Hotspots · emerging

KOODAL — Technical Architecture

Citizen Government AI assists



MAIN DATA FLOW



REAL-WORLD USE CASES

Built for everyday India.



Monsoon flooding

Neighbours verify a blocked drain in hours. The ward team gets a target date.



Sewage overflow

A Tamil voice note goes straight to Metrowater.



Potholes

Many reports merge into one case, backed by stronger evidence.



Streetlights

Citizens confirm with before/after photos. If it isn't fixed, the case reopens.

REAL-TIME SCENARIO · VELACHERY, CHENNAI

Divya's report, start to finish.



Citizen
08:10

Logs in

Phone OTP. Anonymous by default if she wants.



Citizen
08:12

Browses the feed

Supports the Phoenix-signal pothole with a press and hold.



Citizen
08:20

Creates a report

Photo + Tamil voice note: sewage on 100 Feet Road.



Gemini
08:21

AI match

Sewage · Critical · Metrowater.
92% match with CP-2107, so she joins it.



Community
11:40

Verified

41 supporters + validations push confidence past 80%.



Government
14:05

Official case

AE Ganesan approves CP-CHN-24781 and assigns Metrowater with a target date.



Government
Day 2

In progress

Crew on site. Updates appear on the timeline and Divya is notified.



Citizens
Day 4

Fixed → Closed

Proof photo posted, 25 citizens confirm and the case closes. If 3 say not fixed, it reopens.

REAL-TIME SCENARIO · GREATER CHENNAI CORP.

An officer's day on KOODAL.



Secure sign-in

Employee ID + password, then OTP.

Officer
09:00



Dashboard

6 pending · 3 overdue · 1 reopened. “Review now.”

Officer
09:02



Triage

The queue is sorted by priority score, with an AI summary of each case's evidence.

Gemini
09:05



Approve / reject

Approve: set department, team and target.
Reject: give a reason citizens can see.

Officer
09:15



Board & deadlines

Drag a card to In progress. Overdue cases are flagged on the public timeline.

Officer
Daily



Mark fixed

Attach a proof photo. Citizens confirm, and only then does the case close.

Officer
Close-out

SOCIAL IMPACT

Trust, on every side.

Citizens

- ✓ A voice in Tamil or English
- ✓ Proof, not promises
- ✓ Safe to report anonymously
- ✓ Live updates on every case they follow

Community

- ✓ A nearby feed of what's happening
- ✓ Neighbours verify what's real
- ✓ Many voices become one strong case
- ✓ Collective sign-off on every fix

Government

- ✓ A verified-first queue, less noise
- ✓ Clear owners and deadlines
- ✓ Evidence-based priorities
- ✓ Public credit for real fixes

STRENGTHS & CHALLENGES

Hard problems, designed for.

Closed loop

Meaningful AI

Working product

Scales city by city

CHALLENGE

HOW KODAL ADDRESSES IT

Fake or spam reports

→ Community verification at 80% confidence, plus an AI duplicate check

Low digital literacy

→ Voice-first · Tamil + English · installable PWA

"Fixed" without fixing

→ Only citizens close cases. 3 "not fixed" votes reopen it.

Department adoption

→ A simple console: approve, assign, mark fixed

Scale across cities

→ Serverless Firebase + Vercel, with data scoped by city

WHAT'S NEXT

More Indian languages

WhatsApp & IVR reporting

Open civic data for cities

Predictive maintenance with Gemini

Pan-India rollout

Report it. Verify it. Resolve it.

KOODAL closes the loop.



TEAM TRIOZ

Thank you

Santhosh S

Team Leader

Aakash T

Team Member